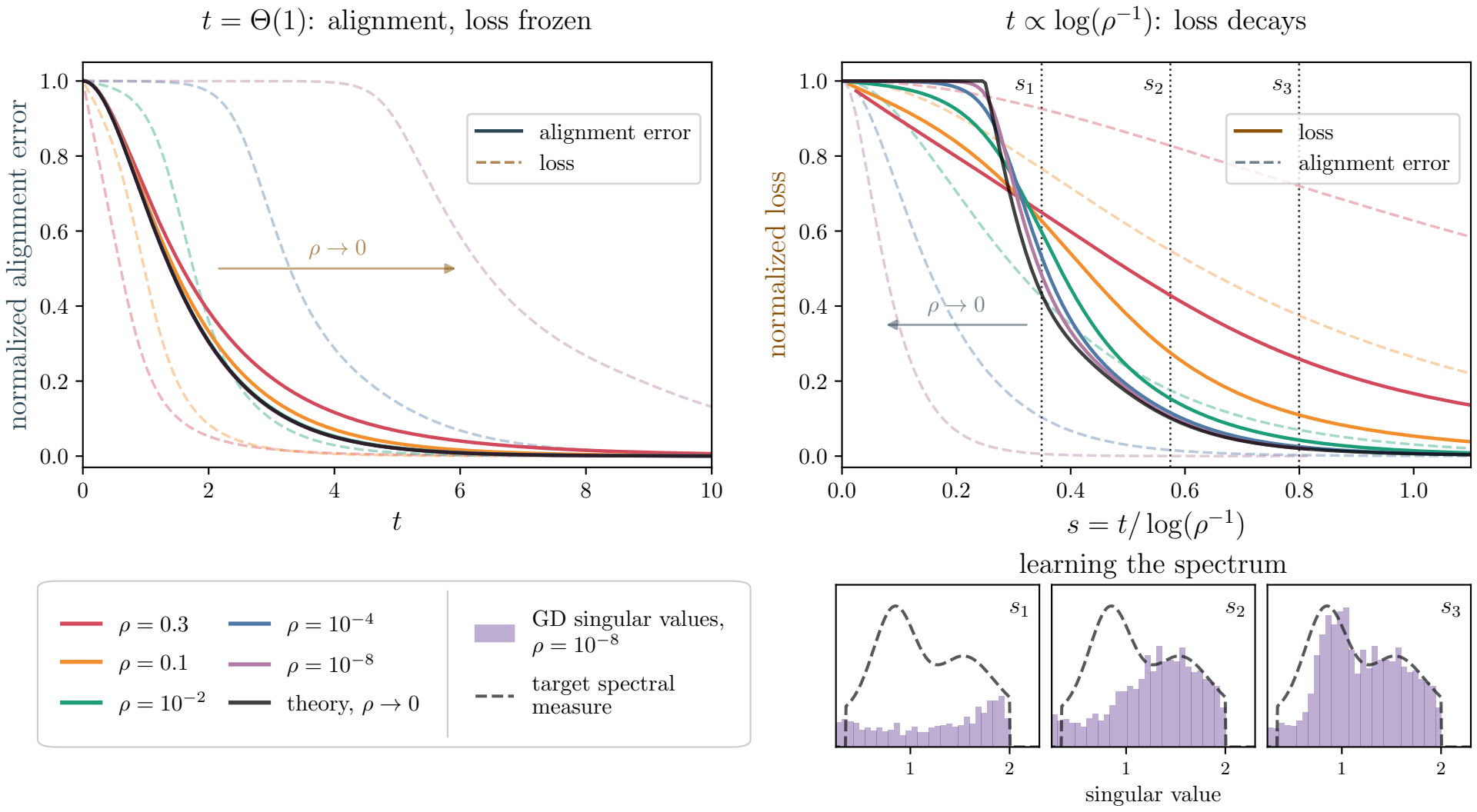

$t = \Theta(1)$: alignment, loss frozen
$t \propto \log(\rho^{-1})$: loss decays
normalized alignment error
alignment error
loss
$\rho \to 0$
t
normalized loss
s_1
s_2
s_3
loss
alignment error
$\rho \to 0$
$s = t/\log(\rho^{-1})$
$\rho = 0.3$
$\rho = 0.1$
$\rho = 10^{-2}$
$\rho = 10^{-4}$
$\rho = 10^{-8}$
theory, $\rho \to 0$
GD singular values, $\rho = 10^{-8}$
target spectral measure
learning the spectrum
singular value